Geographic Information Retrieval (2026W)

Yingjing Huang & Zilong Liu
yingjing.huang@univie.ac.at
zilong.liu@univie.ac.at

Lecture 2

Geoparsing I

Learning Objectives

- ❏ Part I: Why
- ❏ Part II: What
- ❏ **Part III: How**

Why Geoparsing?

Text in GIR

- **Query**: “Places in Austria”
- **Document**: “I had lunch at the Cafe Latte Art, which is in the 9th district and on Universitätsstraße. Right across the street you will find the NIG, where we are having the class today.”
- The **GIR system** need to know where a query or a document is about.

Query -> GIR System -> Documents

What is Geoparsing?

Geoparsing in a Nutshell

- **Definition**: identifying references to locations as expressed in natural language text
- **Alternative name**: “toponym recognition”
- “Toponym”: place name

Geoparsing -> NER

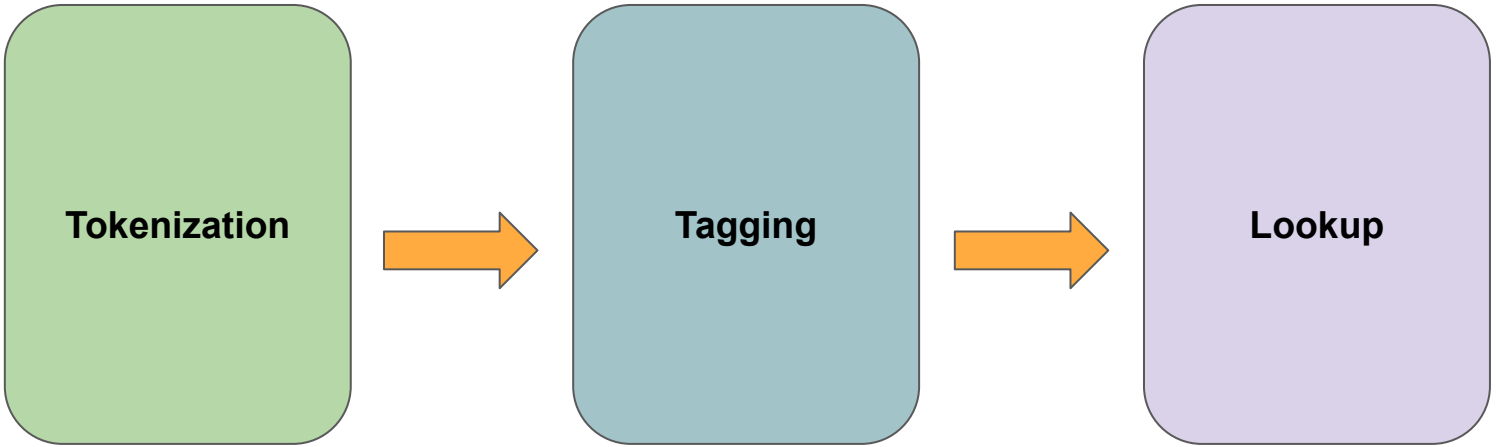
- **Another definition**: a subtask of Named Entity Recognition (NER).
- **NER**: locating and classifying named entities mentioned in text
- **Named entities**: person, date, location, …



How to Geoparse

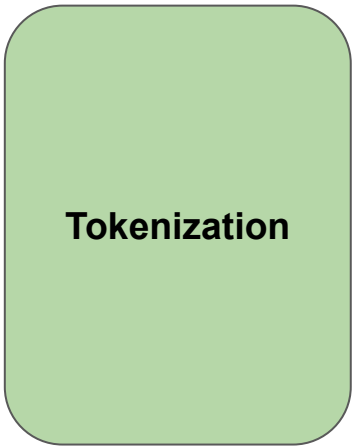
How to Geoparse -> How to NER

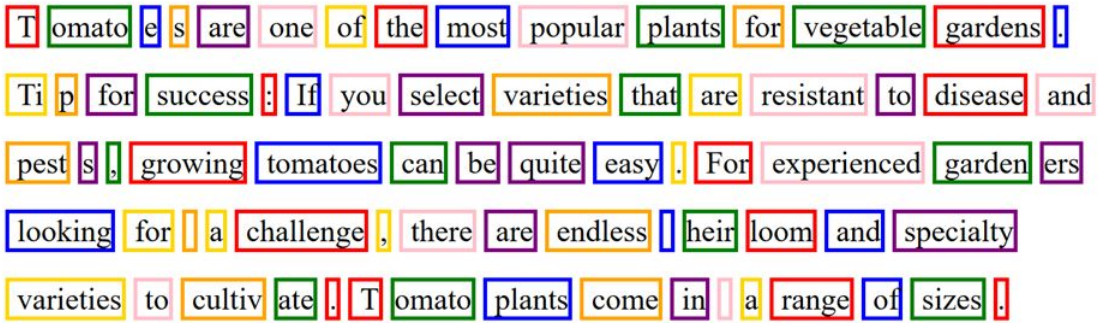
- The NER pipeline consists of three stages.



The NER Pipeline

- **Tokenization**: splitting text into tokens
- **Token**: words, sub-words, and sometimes even characters



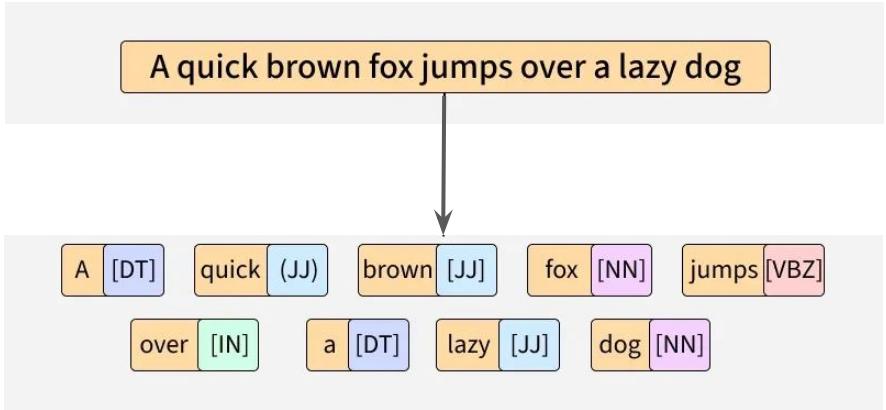


https://www.ibm.com/think/topics/tokenization

The NER Pipeline (Cont'd)

- **Tagging**: assigning a part-of-speech (PoS) label to each token
- **Alternative name**: "PoS tagging"
- **Part of speech**: grammatical categories (e.g., noun, verb, adjective)

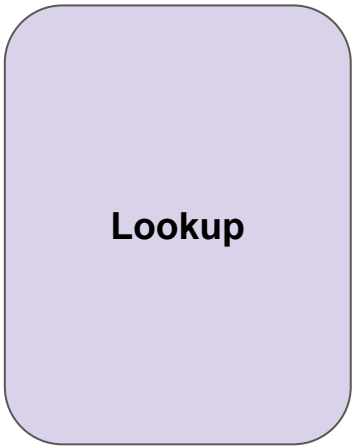




https://www.geeksforgeeks.org/nlp/nlp-part-of-speech-default-tagging/

The NER Pipeline (Cont'd)

- **Lookup**: to look up lists of known locations, people, etc.



Austria *geographical name*

Aus·tria 'ȯ-strē-ə 'ä-

variants *or German* **Österreich** 'œ-stər-ˌrīk̲ 'œ-stə-

landlocked country of central Europe in and north of the eastern Alps mountain system with the Danube River crossing it in the north; a republic; capital Vienna *area* 32,383 square miles (83,871 square kilometers)

https://www.merriam-webster.com/dictionary/austria

How to NER -> How to Geoparse

- The lookup stage is where NER turns into geoparsing.
- Where can we look up for place names?
- **Gazetteer**: *geographical* dictionary

GeoNames: An Example Gazetteer

- **Link**: https://www.geonames.org/
- Let's try changing the selection of "all countries" to "Austria".



Problems with Lookup

- **Finiteness of the list**
  - Gazetteers may not always be up-to-update.
- **Place name variants**
  - Gazetteers may not store “Wien” for Vienna.
- **Exact/fuzzy matching**
  - Gazetteers may not be able to match “Vienn a” to “Vienna”.

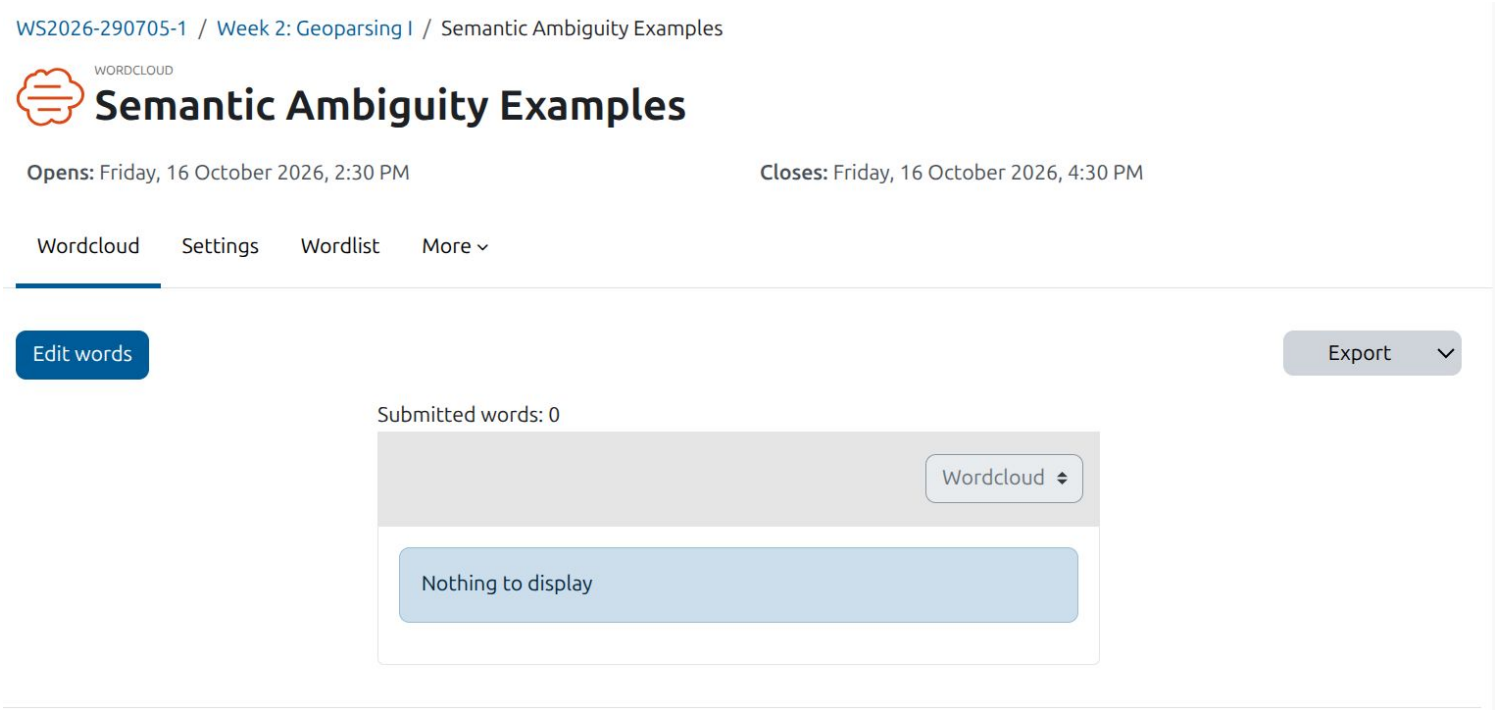
A Bigger Problem with Tagging + Lookup

- **Semantic ambiguity**
  - “Baden” - a named entity, as it’s a town in Germany, Austria, …
  - “Baden” - a verb meaning *to bathe* in German





More Examples of Semantic Ambiguity?



How to Geoparse (in Other Ways)

Tokenization + …

- First, we **keep** tokenization.
- Second, we **replace** tagging + lookup.

Rule-based Methods

- A **rule** can be used to define and express the surrounding **context** of a word.

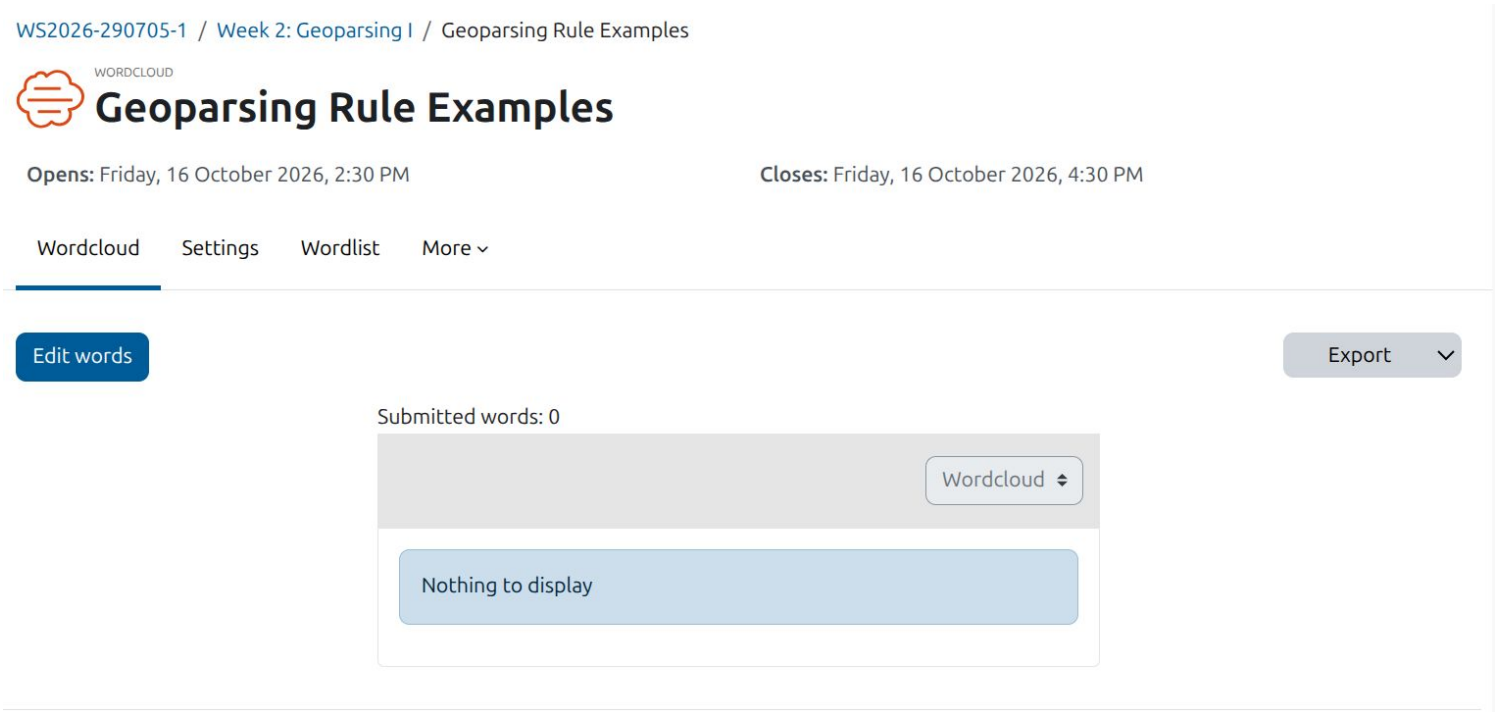
"Sunset Boulevard"

"North of Vienna"

Rule-based Methods (Cont'd)

- Using a specified rule, two kinds of contextual evidence can be captured.
  - **Internal evidence**
    - Example rule: `CapWord + Street/Boulevard/Avenue/Road`
    - Example word matching this pattern: "Sunset Boulevard"
  - **External evidence**
    - Example rule: `North of CapWord`
    - Example word matching this pattern: "North of Vienna"

More Examples of Geoparsing Rules?

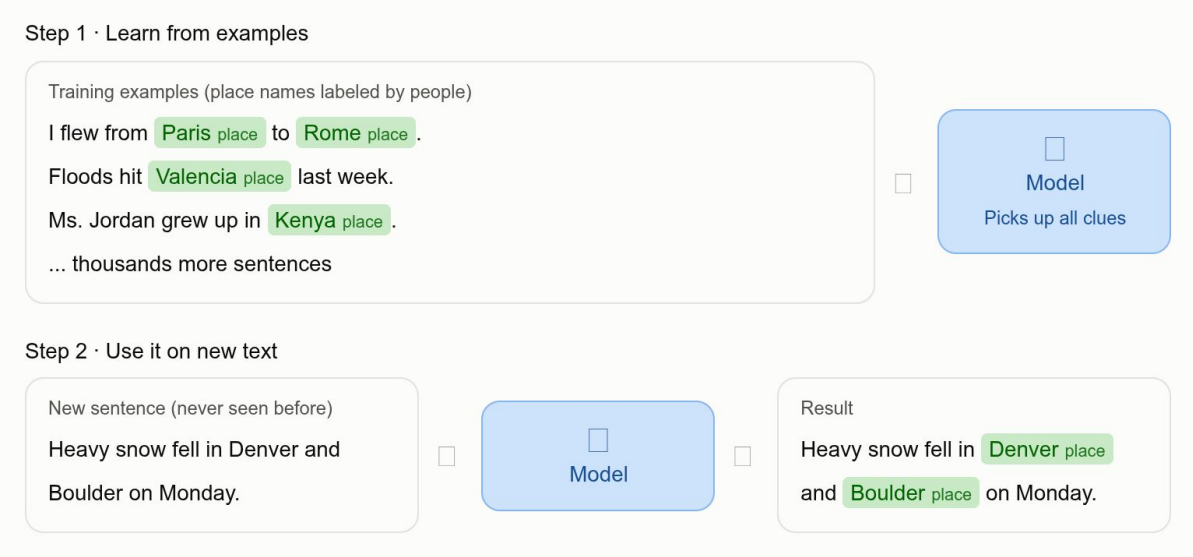


A Problem with Rules

- Rules are **hand-crafted**.
- What if a rule is not specified but a pattern needs to be identified?

Machine Learning–based Methods

- First, a model is trained on an **annotated** text dataset.
- **Rules** are learned implicitly as the model sees both **place names and their context**.
- Then, the model can achieve geoparsing on **unseen** text.



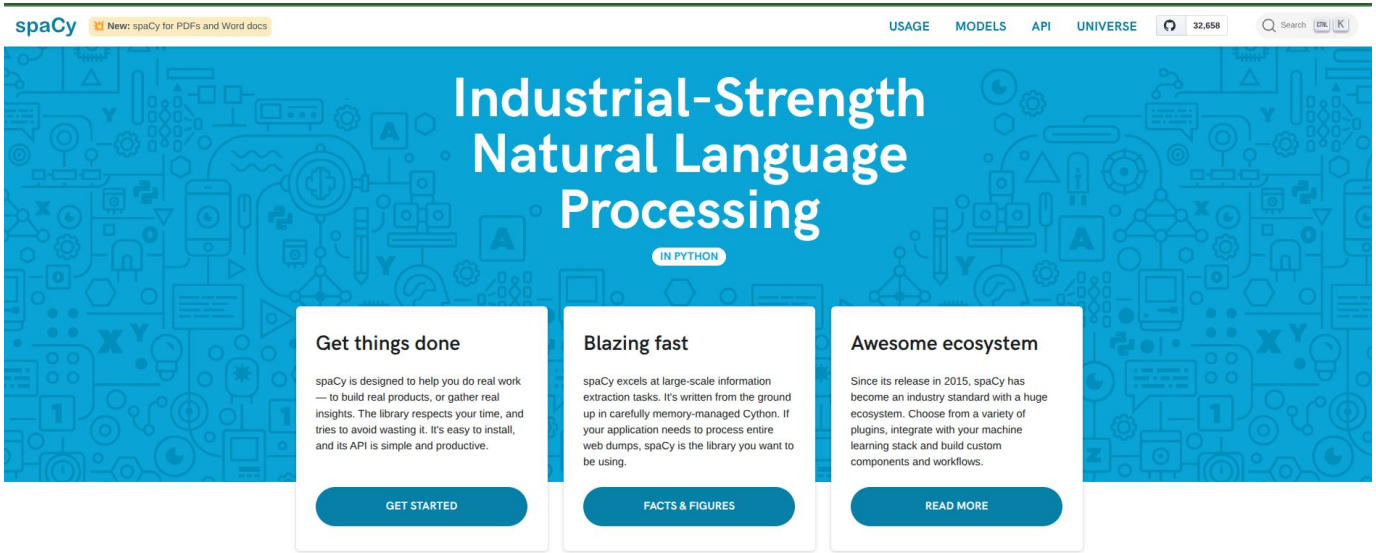
Machine Learning–based Methods

- Do we need to train models in this class?
- No, we can directly use pre-trained models.
- These models have learned knowledge from **a large amount of data**.

How to Geoparse (in a Modern Way)

spaCy: a Machine Learning–based Python Package

- **Link**: https://spacy.io



spaCy - Tokenization

Editable Code — spaCy v3.7 · Python 3 · via Binder

```
import spacy

nlp = spacy.load("en_core_web_sm")
doc = nlp("Apple is looking at buying U.K. startup for $1 billion")
for token in doc:
    print(token.text)
```

RUN

0	1	2	3	4	5	6	7	8	9	10
Apple	is	looking	at	buying	U.K.	startup	for	$	1	billion

https://spacy.io/usage/spacy-101

spaCy - Trained Pipelines

This is called a trained pipeline, i.e., a collection of models.

Editable Code spaCy v3.7 · Python 3 · via Binder

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en_core_web_sm

RELEASE DETAILS

Latest: 3.8.0

English pipeline optimized for CPU. Components: tok2vec, tagger, parser, senter, ner, attribute_ruler, lemmatizer.

LANGUAGE	EN English
TYPE	CORE Vocabulary, syntax, entities
GENRE	WEB written text (blogs, news, comments)
SIZE	SM 12 MB
COMPONENTS	tok2vec, tagger, parser, senter, attribute_ruler, lemmatizer, ner
PIPELINE	tok2vec, tagger, parser, attribute_ruler, lemmatizer, ner

*The **small** English pipeline*

https://spacy.io/usage/spacy-101

spaCy - NER

Editable Code spaCy v3.7 · Python 3 · via Binder

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import spacy

nlp = spacy.load("en_core_web_sm")
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for ent in doc.ents:
    print(ent.text, ent.start_char, ent.end_char, ent.label_)
```

RUN

TEXT	START	END	LABEL	DESCRIPTION
Apple	0	5	ORG	Companies, agencies, institutions.
U.K.	27	31	GPE	Geopolitical entity, i.e. countries, cities, states.
$1 billion	44	54	MONEY	Monetary values, including unit.

https://spacy.io/usage/spacy-101

spaCy - Geoparsing

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*In spaCy, place names can come from these categories: **NORP, FAC, ORG, GPE,** and **LOC**.*

SpaCy recognizes the following built-in entity types:

PERSON - People, including fictional.

NORP - Nationalities or religious or political groups.

FAC - Buildings, airports, highways, bridges, etc.

ORG - Companies, agencies, institutions, etc.

GPE - Countries, cities, states.

LOC - Non-GPE locations, mountain ranges, bodies of water.

PRODUCT - Objects, vehicles, foods, etc. (Not services.)

EVENT - Named hurricanes, battles, wars, sports events, etc.

WORK_OF_ART - Titles of books, songs, etc.

LAW - Named documents made into laws.

LANGUAGE - Any named language.

DATE - Absolute or relative dates or periods.

TIME - Times smaller than a day.

PERCENT - Percentage, including "%".

MONEY - Monetary values, including unit.

QUANTITY - Measurements, as of weight or distance.

ORDINAL - "first", "second", etc.

CARDINAL - Numerals that do not fall under another type.

https://spacy.io/usage/spacy-101

spaCy - Visualizers

Visualizing the entity recognizer

The entity visualizer, `ent`, highlights named entities and their labels in a text.

NAMED ENTITY EXAMPLE

```
import spacy
from spacy import displacy

text = "When Sebastian Thrun started working on self-driving cars at Google in 2007, few people outside of the company

nlp = spacy.load("en_core_web_sm")
doc = nlp(text)
displacy.serve(doc, style="ent")
```



https://spacy.io/usage/visualizers

Summary

- To match a query with a document for GIR, we need to **extract place names from text**.
- Geoparsing is the task designed for this purpose and is a **subtask of NER**.
- Among the various ways of geoparsing, **machine learning**–based ones work the best.
- **spaCy** is one of the most widely used Python libraries for NER, and therefore, geoparsing.

Announcement

Recommended Reading

- **Section 1 (Page 1 - 7) and Section 4.3 (Page 34- 37)** of *Geographic information retrieval: Progress and challenges in spatial search of text*

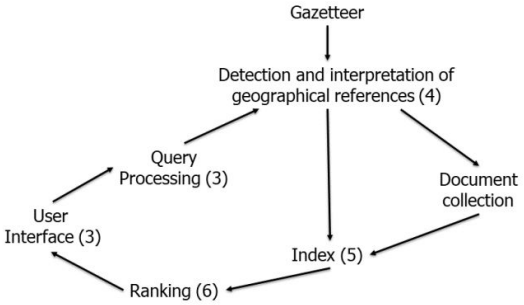


Figure 1.1: Schematic conceptual diagram of a GIR system and its related components - numbers indicate chapters where these topics are handled

Table 4.1: Example output from identifying named entities.

Input: French company Aeroports de Paris International is the design consultant for the work. The company also operates Charles de Gaulle airport in Paris, where part of a terminal building's roof collapsed in May, killing four people.
Output: French company `<org>`Aeroports de Paris International`</org>` is the design consultant for the work. The company also operates `<loc>`Charles de Gaulle airport`</loc>` in `<loc>`Paris`</loc>`, where part of a terminal building's roof collapsed in May, killing four people.

Lab Session

Submission

- Feel free to raise your hands to ask questions.
- Work independently but feel free to discuss with your colleagues and me.
- Make sure to upload your **.ipynb file** to Moodle by 23:59 today for your participation grade.
- In your Jupyter notebook, **include your codes and answers with a clear format**.